Remote Sensing Image Captioning Using Hire-MLP

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***Abstract*—Remote sensing image captioning has been widely applied to traffic management, geographic research, etc. Although the neural network approach has been successfully improving the performance of the Remote sensing image captioning system, Remote sensing image captioning is still facing object identification challenges due to the small size of objects, uneven distribution, and high coupling with the surrounding image background. In this paper, we propose a novel remote sensing image captioning encoder-decoder model Hierarchical rearrangement-Multi-Layer Perceptron (HMLP) whose encoder adapts hierarchical rearrangement-multi-layer perceptron to improve the capability of objection recognition. Extensive experiments have been conducted to testify HMLP by three datasets RSCID, UCM-caption, and NWPU-caption. Results show that HMLP outperforms many image captioning systems in the evaluation metrics BLEU4, METEOR, ROUGE-L, and CIDEr.**

Keywords-Image Captioning; Remote Sensing Image; Multilayer Perceptron; Deep Learning; Computer Vision

I. INTRODUCTION

After years of investigation, artificial intelligence research has achieved significant strides in unimodal fields like computer vision and the understanding and creation of natural language. The goal of cross-modal research is to enable computers to comprehend the semantics of an image and produce natural language to explain that information[1]. Image captioning is a typical application of this study. By using a satellite or an aircraft, remote sensing technology allows us to take pictures and create images of terrestrial landscapes from a distance. The information contained in remote sensing images that were taken using remote sensing technology is described in remote sensing image captioning. Technology for creating remote sensing image captioning is essential in fields like traffic control and geographic study[2].

MLP (Multilayer Perceptron)-only models have emerged as a new development in computer vision in recent years [3, 4]. While removing the strong self-attentive mechanism module, MLP-based models can produce outcomes that are comparable to CNNs. For instance, MLP-Mixer [3] gathers remote information using an MLP applied across patches and extracts information about each place with an MLP applied to each picture patch. The task of remote sensing image captioning first uses the encoder-decoder architecture. With the encoder's image feature extraction networks AlexNet, VGG16, and ResNet[5], existing work in supervised learning has produced positive results, and the decoder produces image captioning using a modified long short-term memory network LSTM(Long Short-Term Memory), and a gated recurrent unit GRU (Gated Recurrent Unit) for optimization. The encoder-decoder model, like in Xu, Lu, Chen, and Wang, is then further optimized by the addition of the self-attentive mechanism[6].

The remote sensing image itself has issues with small size, high density, and uneven distribution of remote sensing targets, which make them difficult to identify and capture, as well as a high degree of coupling with the surrounding environment and a large scale span. For example, the similarity of background color and remote sensing target color makes it challenging to identify remote sensing targets[7,8]. Due to these issues, feature information is not properly utilized inside the network and classic image captioning models struggle to effectively extract image feature information[9]. This paper suggests the HMLP model as a solution to this issue. The HMLP model divides the local information of remote sensing images into regions, rearranges the information within the regions, rearranges the information across regions, and further obtains more adequate global information to effectively distinguish confusable objects, thus obtaining better remote sensing image features. The effectiveness of this study's method is tested on three datasets, namely RSCID[6], UCM-caption[5], and

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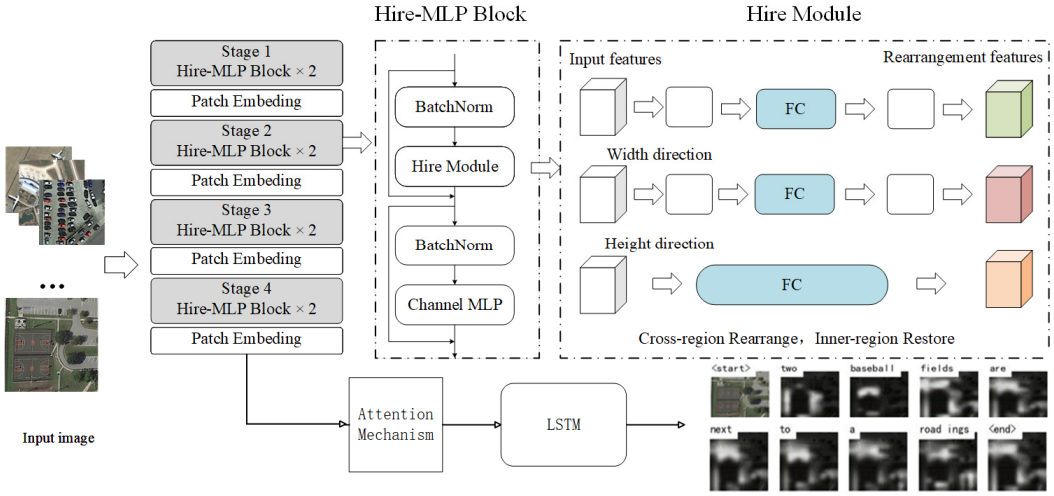


Figure 1: HMLP remote sensing image captioning generation model

NWPU-caption[10]. In this paper, Hire-MLP(Hierarchical Rearrangement - Multi-Layer Perceptron Structure) is used for the feature extraction step in remote sensing image captioning generation to conduct acceptable extraction of remote sensing image features.

II. HMLP

The proposed HMLP remote sensing image captioning generation model is described in this part, and Fig.1 depicts the model's overall structure. The model uses the conventional image encoder-sentence decoder framework to generate description statements and extract picture features. To fully extract the spatial information, environmental information, and scene information from the remote sensing image, the Hire-MLP structure is introduced during the image feature extraction stage. It performs cross-region rearrangement based on the local information captured in the spatial region of the remote sensing image to capture the global contextual information of the remote sensing image. Fully extracted from the complex image elements include scene information, environmental information, and geographical information. The attention signal is used to direct the generation of remote-sensing image captioning statements, and the attention mechanism is also introduced in the generation stage of remote-sensing image captioning to focus on the image information and associate the most pertinent words in the generated region.

A. Encoders – Decoders

Modern results can be achieved by directly optimizing the likelihood that a particular input sentence will be translated correctly in an "end-to-end" manner, according to advancements in the creation of remote sensing image captioning. These models 'decode' the vectors into the appropriate output phrases by using recurrent neural networks to encode variable-length inputs into fixed-dimensional vectors. A hierarchically rearranged visual MLP structure is used in this study to encode the local and global information and to average the fully connected vectors of the new cycle by the spatial movement of the local information in different directions. The Hire-MLP structure is used for feature extraction, and the input remote sensing images are used to map the extracted features into visual space. The attributes of the reorganized remote sensing image are also output.

The captioning generation of remote sensing images can be regarded as a sequence-to-sequence learning task, using the decoder to decode the intermediate vector obtained from the encoder into a sequence of words $Y=(y_1,y_2,...,y_n)$, where $y_i \in D$ is the predicted generated words, D is the lexicon containing all words, T is the maximum length of the utterance words are represented by the word embedding vector, each utterance starts with a special start marker <start> and ends with a special end marker <end>. In the decoding process of the model, the words generated in the previous time step are fed into the LSTM, and the hidden state h_t of the current time step LSTM is generated in combination with the attention mechanism, and then the current word y_t is predicted to be generated based on h_t.

$$v_t = \partial\left(W_{kx}x_t + W_{kh}h_{t-1}\right) \tag{1}$$

$$m_t = \partial\left(W_{mx}x_t + W_{mh}h_{t-1}\right) \tag{2}$$

$$p_t = \partial\left(W_{px}x_t + W_{ph}h_{t-1}\right) \tag{3}$$

$$z_t = m_t \odot z_{t-1} + v_t \odot \tanh\left(W_{zx}x_t + W_{zh}h_{t-1}\right) \tag{4}$$

$$h_t = p_t \odot \tanh\left(z_t\right) \tag{5}$$

where ∂ is the sigmoid function; z is the memory cell, m is the forgetting gate, v is the input gate and p is the output gate.

B. Hire-MLP

Some crucial local information might go unnoticed since hybrid operations capture and aggregate global information while classic MLP approaches have a fixed number of dimensions in the complete connectivity layer. Better local and

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global information is captured in remote sensing photos with the introduction of the Hire-MLP module. The Hire-MLP module divides, rearranges within regions, and rearranges across regions using local information from remote sensing photos, improving the features of those images. The Hire-MLP architecture is composed of several Hire-MLP blocks stacked on top of one another. Each Hire-MLP block is divided into two smaller blocks, the Hire module and the channel MLP, which combine spatial and channel data, respectively. The Hire-MLP block can be shown as, given an input feature $X \in R^{H \times W \times C}$, its height H, width W, and channel number C.

$$Y = Hire(\mathrm{BN(X)}) + \mathrm{X} \tag{6}$$

$$Z = ChannelMLP(\mathrm{BN(Y)}) + Y \tag{7}$$

where BN stands for batch normalization, Y and Z stand for the module's intermediate and output features, respectively, and the entire Hire-MLP architecture is an iterative stacking of Hire-MLP blocks for global information capture.

The input features are split into multiple regions, where they are then rearranged intra-regionally in each region (the input features may be split along the width or height direction) before cross-region rearrangement operations are used to obtain the global feature information. For instance, the input feature X of the form $H \times W \times C$ is separated into g areas, $X = [X_1, X_2, ..., \mathrm{X}_g]$, and the remote sensing image features are intra-regionally segmented in the height direction. There are h tokens in each zone along the height direction, where $h = h / g$. then the traits are intra-regionally reorganized. All tokens in X are joined together along the channel dimensions, starting with the input feature $X_i \in R^{H \times W \times C}$ at the ith area along the height direction, to create a rearranged feature X_i of shape $W \times hC$. Send X_i to the MLP module F's fully connected layer to mix the data along the final dimension and create the output features. The model may then collect global spatial information by moving the tokens along the height/width direction and sharing information between various regions. This is done by performing a cross-region rearrangement operation.

III. EXPERIMENT

A. Datasets

In this study, three datasets—RSCID, UCM-caption, and the most recent NWPU-caption—are used to generate remote sensing image captioning with NWPU captions serving as the largest dataset of its kind. RSICD dataset, as suggested by et al. The dataset's picture sources are SkyMap (Tianditu), Baidu Maps MapABC, and Google Earth [11]. For a total of 10,921 photos across 30 categories, this collection includes image caption pairs. Dataset for UCM-Captions: There are 100 photos of each of the 21 types of land use. The 157,500 remote-sensing image captioning statements and 31,500 images in the NWPU-Captions collection were individually hand-annotated by seven skilled volunteers.

The commonly used BLEU1, BLEU4[12], CIDEr[13], METEOR[14], and ROUGE[15] metrics were utilized to gauge the effectiveness of the utterance of the generated remote sensing image captioning to ensure the comparability of the approaches in this study.

B. Experimental setup

The baseline model in this paper, AlexNet+LSTM[16], as well as the usual approaches for generating remote sensing image captioning (Soft [6], Hard [6], Mlca [17], SCST [18], and UpDown [19]) are used to evaluate the model described in this paper on three distinct datasets. These are the precise implementation and test parameters: The techniques in this paper were implemented using the Torch-based PyTorch machine learning toolkit, which was created by the FAIR (Facebook Artificial Intelligence Research Institute). Dropout was set to 0.5, the Adam optimization technique was used to optimize the model parameters, 16 was chosen for the batch training size, 1e-4 was chosen for the encoder learning rate, and 4e-4 was chosen for the decoder learning rate to prevent overfitting.

C. Analysis of experimental results

The HMLP model in this article performs well on all three remote sensing image captioning datasets, as shown in Tables 1, 2, and 3. It outperforms the other models in CIDEr measures on the RSCID and UCM-captions datasets and all evaluation metrics on the biggest scale NWPU-captions dataset. Overall, the model performs better when it comes to creating remote-sensing image captioning.

Table 1. Results of NWPU-caption

Model	BLEU1	BLEU4	METEOR	ROUGE	CIDEr
Baseline[16]	0.717	0.440	0.320	0.578	1.065
Soft[6]	0.731	0.455	0.339	0.599	1.179
Hard[6]	0.733	0.464	0.340	0.600	1.103
SCST[18]	0.735	0.471	0.330	0.589	1.149
UpDown[19]	0.738	0.472	0.338	0.600	1.140
MLCA[17]	0.745	0.478	0.337	0.601	1.264
Our	**0.789**	**0.511**	**0.351**	**0.633**	**1.301**

Table 2. Results of RSCID

Model	BLEU1	BLEU4	METEOR	ROUGE	CIDEr
Baseline[16]	0.579	0.259	0.261	0.529	2.052
Soft[6]	0.656	0.344	0.329	0.610	1.874
Hard[6]	0.689	0.368	0.335	0.626	1.983
MLCA[17]	0.750	0.459	0.342	0.638	2.418
Our	0.736	0.413	0.330	0.624	**2.432**

Table 3. Results of UCM-Captions

Model	BLEU1	BLEU4	METEOR	ROUGE	CIDEr
Baseline[16]	0.701	0.503	0.346	0.651	2.313
Soft[6]	0.796	0.598	0.416	0.749	2.823
Hard[6]	0.784	0.601	0.430	0.773	3.016
SCST[18]	0.810	0.657	0.417	0.750	3.054
UpDown[19]	0.823	0.668	0.435	0.763	3.021
MLCA[17]	0.826	0.668	0.435	0.772	3.240
Our	0.847	**0.716**	0.445	**0.817**	**3.360**

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IV. Conclusions

This paper has proposed a remote sensing image captioning model based on Hierarchical rearrangement-Multi-Layer Perceptron (HMLP) that aims to optimize the capability of local and global visual features extraction of remote sensing images. The experimental findings demonstrate that the HMLP model is capable of significantly enhancing the quality of the creation of remote-sensing image captioning. This model ensures the accuracy of object recognition by not missing any crucial information in remote sensing images.

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